UNIVERSITY of INFORMATION TECHNOLOGY

GRIDSIM

Exploring GridSim Toolkit

GROUP-I

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1. INTRODUCTION

1.1. GRID COMPUTING

The widespread use of the Internet and the availability of powerful computers and fast networks have changed the way we perform large-scale parallel and distributed computing. This includes connecting computers and resources from different locations to solve complex problems. This approach is known as "grid computing" and "peer-to-peer (P2P) computing." These networks allow us to share, choose, and combine computational and data resources to solve large-scale data-intensive problems in fields like science, engineering, and commerce.

1.2 GRID COMPUTING LAYERS

1. **Fabric**: This layer includes various types of computers, networks, scientific instruments, and the systems that manage these resources. It can range from small computers to powerful clusters.
2. **Core Middleware**: This layer provides essential services that allow users to access remote resources securely and seamlessly. These services include managing security, handling remote job submissions, managing storage, and providing information about available resources.
3. **User-Level Middleware**: Above the core middleware, there's another layer that offers more user-friendly tools. These tools help users find the right resources, develop applications, and create an adaptive environment for running their programs.

1. INTRODUCTION

1.2. GRID COMPUTING LAYERS (CONTINUED)

Applications: At the top of the stack are the actual applications that people use. These can be programs built using grid-specific libraries or existing applications that have been adapted to work in a grid environment.

1.3. GRID RESOURCE BROKER

A Grid resource broker is the arbiter for access to a Grid's computational resources and therefore its performance and functionality has a wide-ranging influence on the utilization and performance of the Grid. A grid resource broker serves as a critical component in the field of grid computing. Its primary purpose is to efficiently manage and optimize the allocation of computing resources in a grid environment, ensuring that tasks and jobs submitted by users or applications are executed effectively. However, The ability to experiment with a large number of grid scenarios was limited by the number of resources that were available in the WWG. Eg. Nimrod-G.

2. BACKGROUND

2.1. GRIDSIM TOOLKIT

GridSim is a Java-based grid simulation toolkit for the Modeling and Simulation of Distributed Resource Management and Scheduling for Grid Computing. It is a simple framework for deterministic modeling and simulation of resources, network connectivity with different capabilities, configurations, and domain sand applications to evaluate scheduling strategies. It supports primitives for application composition, information services for resource discovery, and interfaces for assigning application tasks to resources and managing their execution

2.2. GRIDSIM FEATURE

- **Resource Modeling**: GridSim allows users to model a wide variety of computing resources, including CPUs, storage devices, data centers, and networks.
- **Network Modeling**: It provides detailed network modeling to simulate network latency, bandwidth, and communication between grid resources.
- **Task and Job Modeling**: Users can model tasks and jobs to simulate the execution of applications and workloads in a grid environment. GridSim supports parallel and distributed computing models, making it suitable for a wide range of applications.
- **Scheduling**: GridSim offers various scheduling algorithms that can be used to allocate tasks to grid resources.

2. BACKGROUND

2.1. GRIDSIM FEATURE(CONTINUED)

- **Fault Tolerance:** GridSim includes mechanisms for simulating resource failures and network interruptions. Researchers can study the impact of such failures on grid application performance and develop fault-tolerant strategies.
- **Workload Generation**: Users can generate synthetic workloads to simulate the behavior of grid applications.
- **Visualization**: GridSim provides visualization tools that help users understand and analyze simulation results.

3. OBJECTIVES

The GridSim Toolkit is a widely-used resource for modeling and simulating distributed and grid computing environments. Its versatile features allow researchers and developers to create realistic simulations that mirror real-world grid scenarios. For instance, when simulating a global grid for climate modeling, the toolkit's capabilities shine, enabling the accurate modeling of distributed resources, network behaviors, and complex computational workloads. Through our testing project, we aim to evaluate the toolkit's performance, scalability, and adaptability in such real-world grid scenarios. Our findings will contribute valuable insights to understand the toolkit's capabilities for more simulations related with grid computing environment.

4. INSTALLATION

1. Prerequisites

- Before installing GridSim, ensure that your system meets the necessary prerequisites, which typically include Java Development Kit (JDK) and a compatible operating system.

2. Download GridSim and Install

- Visit the official GridSim website or repository to download the latest version of the toolkit. You can also download from third party websites like sourceforge.net.

4. Set Environment Variables (Optional):

- Set CLASSPATH or GRIDSIM_HOME, to ensure that Java can locate the necessary GridSim libraries and resources.

```
C:\Program Files\GridSim\bin
```

5. Add GridSim to Path:

- Adding GridSim to your system's PATH allows you to run GridSim commands and scripts from any directory without specifying the full path to the GridSim executable.

6. Run the Application

- You do not need to compile GridSim source code. The JAR file is provided to compile and to run GridSim applications.
- Compile the Java Source file:

```
PS C:\Program Files\GridSim\examples> javac -cp ..\jars\gridsim.jar .\gridsim\example01\Example1.java
```

- Run the Java Class file for result:

```
PS C:\Program Files\GridSim\examples> java -cp ..\jars\gridsim.jar .\gridsim\example01\Example1.java
```

5. TEST PLAN AND RESULTS

5.1. BASIC SIMULATIONS EXAMPLES

Example 1: In this example, we demonstrate how to create a simple Grid resource using the GridSim package. The resource is composed of three machines, each with multiple Processing Elements (PEs) or CPUs.

```
Starting example of how to create one Grid resource
Initializing GridSim package
Initialising...
Starting to create one Grid resource with 3 Machines ...
Creates a Machine list
Creates the 1st Machine that has 4 PEs and stores it into the Machine list
Creates the 2nd Machine that has 4 PEs and stores it into the Machine list
Creates the 3rd Machine that has 2 PEs and stores it into the Machine list

Creates the properties of a Grid resource and stores the Machine list
Finally, creates one Grid resource and stores the properties of a Grid resource
Finish the 1st example
PS C:\Program Files\GridSim\examples>
```

Example 2: In this example, we demonstrate how to create Grid users and Gridlets using the GridSim package.

```
Starting example of how to create Grid users

Creating 8 Gridlets
Creating 3 Grid users

Gridlet ID    User ID    length     file size    output size
    0            0        3500        300         300
    1            0        5000        500         500
    2            0        9000        900         900
    3            1        24314        92         327
    4            1        26558        143         326
    5            1        29487        99         269
    6            2        20391        128         363
    7            2        12326        110         259
Finish the example
```

5. TEST PLAN AND RESULTS

5.2.1 TEST SCENARIO 1

Objective: The objective of this test plan is to comprehensively evaluate the DataGrid example provided within the GridSim toolkit. We aim to assess prototype of its functionality, performance, and the accuracy of data management and replication in a simulated grid environment**.**

Test Scenario 1: In this scenario, we will simulate a DataGrid environment with the following steps:

- **Initialize the Simulation**: Initialize the GridSim simulation environment to set the stage for the DataGrid simulation.
- **Create a Top-Level Replica Catalog**: Instantiate a top-level replica catalog within the DataGrid simulation. This catalog will manage metadata and the location of data replicas.
- **Create Data Grid Resources**: Create data grid resources, which represent nodes or servers in the grid environment. These resources will store and manage data replicas.
- **Create Users**: Generate user entities that will interact with the DataGrid. These users may request and manipulate data objects in the grid.
- **Create Network**: Create a simulated network environment that connects the data grid resources and users. The replica catalog are connected to the network using the attachHost method.
- **Connect Entities to the Network:** Connect the data grid resources and user entities to the simulated network. Ensure that network connectivity is established correctly.

5. TEST PLAN AND RESULTS

5.2.1. TEST SCENARIO 1 - RESULT

```
Initialising...
Top Level Replica Catalogue created!

Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_0 - id: 10)

Resource 0 created.

Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_1 - id: 16)

Resource 1 created.

Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_2 - id: 22)

Resource 2 created.
Dummy user created
Network created
Starting GridSim version 5.0
Entities started.
Dummy_User idling ....
Dummy_User: %%%% Exiting body() at time 0.0
DataGIS: Notify all replica catalog entities for shutting down.
DataGIS: Notify all GridSim entities for shutting down.
Sim_system: No more future events
Gathering simulation data.
Simulation completed.
```

5.2.2. TEST SCENARIO 2

In this scenario, we will simulate a DataGrid environment with FileUser class to add more customized methods the following steps:

- All initial steps are same with Scenario 1.
- **Create Users**: Generate user entities that will interact with the DataGrid. These users may request and manipulate data objects in the grid which represents a user interacting with the DataGrid. The behavior of this user is defined in the FileUser class's body() method.
- How it works
  - Waiting for all entities to register for period
  - Creating three new files and adding them to a specific resource (Res_2).
  - Getting the full name of one of the files (file_0) and its location.
  - Transferring the file from its location to the user entity.
  - Replicating the file to two other resources (Res_0 and Res_1).
  - Attempting to delete a replica of the file from Res_0.
  - Attempting to delete the master file from Res_2.
  - Deleting the replica of the file from Res_1.
  - Finally, deleting the master file from Res_2.

5.2.2. TEST SCENARIO 2 - RESULT

```
Initialising...

Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_0 - id: 10)


Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_1 - id: 16)


Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_2 - id: 22)

Creating a grid user entity with name = File_User, and id = 27
Starting GridSim version 5.0
Entities started.
File_User.addMaster(): file_01 has been added to Res_2
File_User.addMaster(): file_12 has been added to Res_2
File_User.addMaster(): file_23 has been added to Res_2
user:File_User:-Transfer of file file_01 succesful
File_User.replicateFile(): file_01 has been replicated to Res_0
File_User.replicateFile(): file_01 has been replicated to Res_1
File_User.deleteFile(): file_01 has been deleted from Res_0
File_User.deleteFile(): There was an error in deleting file_01 from Res_2
File_User.deleteMaster(): There was an error in deleting file_01 from Res_2
File_User.deleteFile(): file_01 has been deleted from Res_1
File_User.deleteMaster(): file_01 has been deleted from Res_2
File_User:%%%% Exiting body() at time 577.3651999997123
DataGIS: Notify all replica catalog entities for shutting down.
DataGIS: Notify all GridSim entities for shutting down.
Sim_system: No more future events
Gathering simulation data.
Simulation completed.
PS C:\Program Files\GridSim\examples>
```

5.2.3. TEST SCENARIO 3

In this scenario, we will simulate a DataGrid environment with FileUser class to add more customized methods the following steps:

- All initial steps are same with Scenario 1.
- **Create User**: Create a DataGridletUser (user) with specific network parameters. This user is responsible for handling DataGridlets.
- **Create Network**: Create two routers (r1 and r2) and establish a network link (link) between them with certain network parameters.
  - Attach schedulers to routers.
  - Attach r2 to r1 using the link.
- **Connect Resources, Users, and Catalogues to the Network**:
  - Connect the DataGridResources to router r2.
  - Connect the DataGridletUser (user) to router r1.
  - Connect the replica catalogue (rc) to router r2.
- **How it works**
  - Creates two new files (f1 and f2) with specified characteristics.
  - Adds these files as master files to specific resources (Res_0 and Res_1).
  - Creates three DataGridlets with unique IDs, gridlet length, and required files.
  - Submits these DataGridlets to the corresponding resources.
  - Waits for the gridlets to finish execution and receives them back.

5.2.3. TEST SCENARIO 3 - RESULT

```
Initialising...

Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_0 - id: 10)


Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_1 - id: 16)


Starting to create one Grid resource with 3 Machines
Finally, creates one Grid resource (name: Res_2 - id: 22)

Creating a Gridlet user entity with name = Gridlet_User, and id = 27
Starting GridSim version 5.0
Entities started.
Gridlet_User.addMaster(): file11 has been added to Res_0
Gridlet_User.addMaster(): file22 has been added to Res_1
Submitting gridlet 0 to resource 0
Submitting gridlet 1 to resource 1
Submitting gridlet 2 to resource 2
Received back gridlet 2
Received back gridlet 0
Received back gridlet 1
Gridlet_User:%%%% Exiting body() at time 182578.99051989397
DataGIS: Notify all replica catalog entities for shutting down.
DataGIS: Notify all GridSim entities for shutting down.
Sim_system: No more future events
Gathering simulation data.
Simulation completed.
PS C:\Program Files\GridSim\examples>
```

6. CONCLUSION

In this comprehensive exploration of Grid Toolkit Testing, we have DIVED into the world of modeling, simulating, and evaluating grid and distributed computing environments. Throughout this journey, we have uncovered the fundamental concepts, techniques, and tools necessary for assessing the performance, reliability, and scalability of grid-based systems. From harnessing the power of GridSim to crafting intricate simulations, we have witnessed how this toolkit empowers researchers, engineers, and developers to gain invaluable insights into the behavior of complex distributed systems.

7. REFERENCES

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